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CS 654 MACHINE LEARNING AND DATA MINING

Introduction

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WHAT IS MACHINE LEARNING?

- Machine Learning is a branch of Artificial Intelligence (AI).
- Machine Learning imitates how humans learn.
- Many Machine Learning algorithms have been successfully applied in solving Data Mining/Data Science problems.
- Two major areas in Machine Learning are
 - Supervised learning
 - Unsupervised Learning

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WHAT IS DATA MINING?

- Extraction of interesting and useful patterns from data.
- Common alternative names:
 - Knowledge **D**iscovery from **D**ata, knowledge extraction, data/pattern analysis, business intelligence (BI)
- Data Mining often uses Machine Learning methods to mine data for patterns.

MAJOR MACHINE LEARNING AREAS

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

SUPERVISED LEARNING

- Learn from data with labels.
- Data mining application
 - Data classification

UNSUPERVISED LEARNING

- Learn patterns from data without labels
- Data mining application
 - Data clustering
 - Association rule mining

REINFORCEMENT LEARNING

- Learn to achieve from interacting with coach and environment.
- Application
 - Play games

MAJOR DATA MINING TASKS

- Data clustering
- Data classification
- Association Rule mining

DATA CLUSTERING

- Put data into groups such that
 - Data within same group are more similar
 - Data within different groups are less similar

DATA CLASSIFICATION

- Use data with labels to train a classifier first.
- Use the trained classifier to classify new data into a known category.
- Binary classification
- Multiclass classification

ASSOCIATION RULE MINING

- Basket Analysis
- Given baskets with purchased items
- Identify which items are likely been purchased together

MACHINE LEARNING PROCESS

- Prepare data
- Train the model
- Validate the model during training
- Test the model after model been constructed

DATA MINING PROCESS

- Pre-mining: Data preparation
- Data mining tasks: clustering, classification, ...
- Post-mining: Pattern evaluation and interpretation



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